



MedVet Silicon Valley
7090 Santa Teresa Boulevard
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06-07-2026

Client:	Thao Nguyen	Patient ID:	645583
Address:	227 Briar Ridge Drive Mc Kuen San Jose, California, 95123	Phone Number:	Thao (Primary): 608-895-4306, Yuheng (secondary): 530-761-6944
Patient:	Bo	Age:	Deceased (05-24-2026)
Species:	Canine	Sex:	MN
Breed:	Shiba Inu	Color:	Tan/White

The following is a copy of the medical records for Bo from his visit on 05-20-2026.

Presenting Problem

Respiratory distress

Master Problems

Glaucoma

Hx debarked

History

Presented for difficulty breathing. Sunday night he started to breathe with a little more effort than normal, worse today.
Mostly outdoor dog.
Current on vaccines.
Will bite per owner

Health Status

Date/Time	Weight (kg)	Temp(°F)	Heart Rate	Respiratory Rate	Pain	CRT	MM	Attitude	Pulse Quality	Respiratory Effort	Notes
05-22-2026 5:00:00pm											Resp. rate = w/o Respiratory Effort = w/o
05-22-2026 4:00:00pm	9.60		120	36	1-2		Pink		Normal	Shallow	Attitude = dull Pain score = 0/4
05-22-2026 3:00:00pm				30						Normal	
05-22-2026 1:00:00pm				24						Shallow	
05-22-2026 12:00:00pm				28						Shallow	
05-22-2026 11:00:00am				24						Shallow	
05-22-2026 10:00:00am	10.20	101.6	110	24	1-2		Pale		Fair	Shallow	Pain score = 2/4 Attitude = dull
05-22-2026 9:00:00am				18	sec		Pink			Normal	
05-22-2026 8:00:00am				18						Normal	
05-22-2026 7:00:00am				16						Normal	

05-22-2026 6:00:00am			18			Sleeping		Normal	
05-22-2026 5:00:00am			24					Normal	
05-22-2026 4:00:00am			30	0/24	Pink	Sleeping		Normal	Heart rate = hold- resting comfortably Heart rate = hold- resting comfortably. ok per DVM Pulse Quality = hold-see above CRT = uto Weight = uto
05-22-2026 3:00:00am			30					Normal	
05-22-2026 2:00:00am			30					Normal	
05-22-2026 1:00:00am			30					Normal	
05-22-2026 12:00:00am			20					Normal	
05-21-2026 11:00:00pm			18						Respiratory Effort = mmv
05-21-2026 10:00:00pm	99.4	80	18	0/24		Sleeping	Normal	Normal	MM = uto CRT = uto Weight = hold, did not want to get up
05-21-2026 9:00:00pm			16			Sleeping		Normal	
05-21-2026 8:00:00pm			16					Normal	
05-21-2026 7:00:00pm			30					Normal	
05-21-2026 6:00:00pm			28					Normal	
05-21-2026 5:00:00pm			28					Normal	
05-21-2026 4:00:00pm	99.5	120	32	1-2 sec	Pink	QAR	Normal	Normal	Weight = 9.9kg Weight = 9.6kg Pain score = 0-1 BP = 124,120,124 size 3 cuff doppler LPL
05-21-2026 3:00:00pm			30					Normal	Resp. rate = procedure Respiratory Effort = procedure
05-21-2026 2:00:00pm									Resp. rate = ct scan Respiratory Effort = ct scan
05-21-2026 1:00:00pm			28					Normal	
05-21-2026 12:50:01pm	101.4	80	28	1-2 sec	Pink				
05-21-2026 12:00:00pm			28					Normal	
05-21-2026 11:00:00am			42					Normal	
05-21-2026 10:00:00am	9.60	101.4	80	1-2 sec	Pink	Sedate	Normal	Normal	Pain score = 0/4 Resp. rate = panting BP = 124, 130, 128 - #3 RPL
05-21-2026 9:00:00am			30					Normal	
05-21-2026 8:00:00am			24					Normal	

05-21-2026 7:00:00am				24					Normal	
05-21-2026 6:00:00am				24					Normal	
05-21-2026 5:00:00am				36					Normal	
05-21-2026 4:00:00am	9.60	100.5	84	30	0/24	QAR	Normal	Laboured Breathing	MM = uto-angry CRT = uto	
05-21-2026 3:00:00am				30				Normal		
05-21-2026 2:00:00am				20		Sleeping		Normal		
05-21-2026 1:00:00am				18				Normal		
05-21-2026 12:00:00am				18		Sleeping		Normal		
05-20-2026 11:00:00pm				18				Normal		
05-20-2026 10:00:00pm		99.9	60	30	0/24 1-2 sec	Pink Sleeping	Normal	Normal	MM = Pink hard to tell due natural MM color	
05-20-2026 9:00:00pm				28				Normal		
05-20-2026 8:00:00pm				20				Normal		
05-20-2026 7:00:00pm								Normal	Resp. rate = 30/ sedated	
05-20-2026 6:00:00pm	10.20	100.6	88	32	1/24	qar	Fair	Normal	Heart rate = 88/ sedated	
05-20-2026 5:00:00pm									Resp. rate = panting Respiratory Effort = panting	
05-20-2026 4:00:00pm				30					Respiratory Effort = 1/5	
05-20-2026 3:00:00pm	10.20	102.2	120	40			Fair		Respiratory Effort = 1-2/5 MM = unable CRT = unable Pain score = 0-1	

Physical Exam

Weight: 10.2 kg
Temp (°F): 102.2 H.R.: 120 bpm R.R.: 40 rpm
C.R.T. : unable
M.M.: light pink
Attitude: QAR
Hydration: <5% dehydrated
B.C.S.: 6/9
P.S.: 0-1/5

EENT: eyes: clear OU; ears clean AU; no nasal discharge, light pink mm, has periodontal disease
LYMPH NODES: No peripheral lymphadenopathy
CARDIOVASCULAR: no murmur, weak synchronous femoral pulses, sinus arrhythmia
RESPIRATORY: mildly increased RR moderately increased RE, short shallow breaths.
INTEGUMENT: No significant findings. No ectoparasites seen.
ABDOMEN: Soft, non-painful. No masses or organomegaly
MUSCULOSKELETAL: Ambulatory x 4, no significant findings
NEUROLOGIC: Intact cranial nerves. No spinal pain or ataxia
UROGENITAL: No significant findings
RECTAL: externally normal

Therapeutic / Procedure

Thoracocentesis

Specifics

Catheter/Needle Size: 20G catheters

Appearance of Fluid: chylous

Volume Removed:

R Side: 300 mL

L Side: None (only mild amount of pleural fluid)

Any additional comments/complications: No complications. Thoracocentesis performed under sedation, right before abdominal scan.

Thoracocentesis

Specifics

Catheter/Needle Size: 20G catheters

Appearance of Fluid: chylous

Volume Removed:

R Side: 180 mL

L Side: None (no fluid on pocus)

Any additional comments/complications: No complications. Thoracocentesis performed under general anesthesia, right before CT-scan.

Anesthesia / Anesthetic Procedure

Specifics

Anesthesia Plan for thoracocentesis followed by CT-scan thorax + lymphangiogram

- Pre-med: butorphanol 0.2 mg/kg IV + midazolam 0.2 mg/kg IV
- Induction: propofol IV to effect (max 4 mg/kg)
- Maintenance: Isoflurane
- System: Open
- IVF: Normosol 3-5 mL/kg/hr

If hypotensive:

- Give fluid bolus 10 mL/kg over 30 min
- If non-responsive to fluid, start dopamine CRI at 5ug/kg/min

Dopamine 40mg/mL: Mix 0.07 mL with 4.9 mL of NaCl 0.9%, run at 5 mL/hr

If severe bradycardia: glycopyrrolate 0.005mg/kg IV -- can give another dose if no great response after 15 min

*Anesthesia Summary**

Procedure: thoracocentesis followed by CT-scan thorax + lymphangiogram

Doctor: Brassard

Anesthesia Tech: SAM/RB

Anesthesia Maintenance: ISO

Induction time: 13:30

ET Tube Size: 4.0

IV Fluids: Norm R @ 30ml/hr

Other medications: Dex SP 0.1mg/kg IV, Flumazenil 0.34ml IV

Notes: See Smartflow for full notes

Upon intubation patient was noted to have a very small airway, multiple ET tubes were tried prior to sizing down to ultimately a 4.0 to secure airway. Arytenoids noted to erythematous, and there was a significant amount of blood. Dr. Brassard intervened to secure airway (with the 4.0), but there was a soft tissue structure right caual to the arytenoids causing the airway narrowing (suspect mass) Patient became hypotensive, started 10ml/kg fluid bolus/30minutes and started Dopamine 5mcg/kg/min
Removed 180ml of chylous fluid from R side of chest

Brief periods of hypotension + bradycardia throughout procedure that resolved quickly. Occasional 1st degree AV block with two or

three instances of 2nd degree AV block during CT/lymphangiogram
Biopsy was taken from mass (nasopharynx?) and patient was given Dex-SP 0.1mg/kg IV for inflammation
Bloody ET tube noted on extubation, intermittently suctioned back of throat Picture attached to chart of ET tube at extubation

Flumazenil 0.38ml IV given IV 25 minutes after extubation due to slow recovery

Patient Rounded To: IE

Extubation time: 14:53

TVI Norm R: 119.7

TVI Dopamine: 13.45

Total Anesthesia Time: 1h21m

Cardiology In House Consult 2

Specifics

Reason for Consultation: Bicavitary effusion

PHYSICAL EXAMINATION:

Subjective: 5/9 body condition score, QAR, oddly seems sedate but when bothered, will become very aggressive and lunges

Objective:

Cardiovascular: HR = 132 bpm

Murmurs: None auscultated

Other heart sounds: None ausculted

Arrhythmias: None

Heart sounds muffled? Slightly

Femoral pulses: Strong and synchronous

Jugular pulse distension: None

Respiratory rate: 36 breaths/min

Respiratory effort: No effort noted

Lung sounds: No crackles or wheezes ausculted

EENT: MM pink, moist, CRT < 2s

Lymph Nodes: No enlarged lymph nodes palpated

Integument: No significant findings

Abdomen: Soft and nonpainful

Musculoskeletal: Ambulatory x 4

ECHOCARDIOGRAM:

Mitral Valve: The leaflets are not thickened with no prolapse of the anterior leaflet and no prolapse of the posterior leaflet. There is trivial mitral regurgitation.

Tricuspid Valve: The leaflets are not thickened with no visible prolapse of the leaflets. There is trivial tricuspid regurgitation.

Aortic Valve: There is no thickening of the aortic valve leaflets. There is no aortic insufficiency.

Ascending Aorta: Normal

Pulmonic Valve: There is no thickening of the pulmonic valve leaflets. There is no pulmonic insufficiency.

Main and Proximal L/R Pulmonary Arteries: Normal

Left Ventricle: There is no thickening of the interventricular septum and the left ventricular free wall. Chamber size is normal with mild to moderately hypocontractile ventricular wall motion.

Right Ventricle: There is subjectively no thickening of the right ventricle walls. Chamber size is subjectively normal.

Left Atrium: There is no left atrial enlargement.

Right Atrium: There is subjectively no right atrial enlargement.

There is no spontaneous echo contrast, thrombi, or mass lesions noted.

The hepatic vessels are not dilated and there is no ascites. There is mild to moderate pleural effusion.

LVOT: The spectral Doppler pattern is laminar.

RVOT: The spectral Doppler pattern is laminar.

Other: No colorflow evidence of ASD, VSD, or PDA.

Detailed measurements available upon request.

ECHOCARDIOGRAPHIC DIAGNOSIS: Structurally normal heart with decreased contractility; Mild to moderate pleural effusion with normal hepatic vessels

SERIAL ECHO MEASUREMENTS (Measured by MM):

----- LA/Ao / IVSd / LVIDd / LVPWd / FS
05-21-2026 1.07 7.89 30.70 6.82 52%

ASSESSMENT:

1. Structurally normal heart with decreased contractility r/o extra-cardiac disease causing myocardial depression (infection, inflammation, neoplasia??) vs myocardial infarction (less likely given age and breed)
2. Mild to moderate pleural effusion with normal hepatic vessels r/o extra-cardiac

ANESTHETIC RECOMMENDATIONS:

Normal intravenous fluid therapy (e.g., starting crystalloid rate of 5 ml/kg/hr) should be tolerated. One reasonable protocol would include opioid-benzodiazepine premedication, Alfaxan or propofol anesthetic induction, and isoflurane anesthetic maintenance. Provide atropine or glycopyrrolate as needed for bradycardia, but do not automatically include in premedication. Avoid high-dose ketamine (> 5mg/kg) or alpha-2 agonists. Monitoring should include serial blood pressure measurement, continuous electrocardiography, and pulse oximetry.

THERAPEUTIC PLAN:

- Consider pimobendan given poor contractility. Consider high fiber diet and Rutin. Treat any other underlying diseases; if none found and no response to high fiber and Rutin, can consider corticosteroids??
- Recheck in 4-6 month(s) for echocardiogram and thoracic radiographs, always sooner if there are problems or concerns. Timing of subsequent rechecks are dependent on patient status at recheck.
- Exercise Recommendations: Please keep Bo moderately exercise restricted. Please do not encourage them to run, jump, or rough-house. Bo may be allowed to continue to climb stairs.
- Dietary Recommendations: There are no cardiac restrictions at this time. Please continue to feed a normal, healthy diet.
- At-Home Monitoring: Signs of heart disease can include increased respiratory rate, effort or coughing (dogs only; cats do not usually cough from heart disease). If you notice any changes in coughing or trouble breathing, especially when at rest, please have your pet seen by a veterinarian at once. You can count your pet's resting respiratory rate per minute (count the # of breaths in 10 seconds and multiply by 6). A normal respiratory rate is typically less than 40 breaths per minute at rest.

Please notify a veterinarian if any of the following occur:

- Difficulty breathing, breathing faster, breathing harder, or breathing with more effort, especially at rest
- Decreased appetite
- Decreased energy
- Lethargy or weakness
- Fainting, seizures
- Inability to walk
- Worsening of clinical signs

Please note that any cardiac medications must be filled by primary veterinarian, ES doctor, or whomever performs the recheck exam.

If owners would like to meet with a Cardiologist to discuss questions or details and would like a Cardiologist to manage their pet's disease chronically, please schedule a direct appointment with the Cardiology Department and request to be placed on the Wait List if your pet needs to be seen as soon as possible

Thoracocentesis Specifics

Sedation with butorphanol 0.3mg/kg IV
local block with lidocaine 0.5ml total dose

18g cath

Catheter/Needle Size: 18g-gauge over-the-needle catheter
Appearance of Fluid: chylous
Volume Removed:
R Side: 375mLs
L Side: 68mLs

Any additional comments/complications: none
Bandage placed over site: no

Post Tap Weight:

RK

Diagnostic Request

- Point of Care Ultrasound (Ambra)

Ambra Health POCUS

- Radiographs Per Region (Ambra)

Ambra Health DX

- Triglycerides, Serum
- Triglycerides, Fluid

IDEXX Reference Laboratory (IDX)

- ProCyte CBC (IVLS)
- Catalyst Chem17 + Lytes (IVLS)

IDEXX VetLab Station (IVLS)

- Snap 4DX+ Heartworm (IVLS)

IDEXX VetLab Station (IVLS)

- Cytology - Body Fluid Analysis (VDX)

VDx Veterinary Diagnostics and Preclinical Research Services

Specifics pleural effusion

- Echocardiogram In-House Initial (Ambra)

Ambra Health EC

- CT Scan (Ambra)

Ambra Health CT

- Endoscopic Images (Ambra)

Ambra Health ES

- Full / Derm Biopsy (VDX)

VDx Veterinary Diagnostics and Preclinical Research Services

- Triglycerides, Fluid

IDEXX Reference Laboratory (IDX)

- CT Scan (Ambra)

Ambra Health CT

- I-Stat Chem 8+ Serial/Recheck (In-House)

In-House Diagnostics

- Coag Dx PT & aPTT Citrate Initial (IVLS)

IDEXX VetLab Station (IVLS)

Diagnostic Result

Full / Derm Biopsy (VDX) (Ref: US798-DR436381)

Outcome

Attached is VDX Report #: 26052697

Owner: Nguyen, Thao, Pet: Bo US798-DR436381

Doctor: Kelly, Acct: 7070

This report has been sent from an automated system which does not accept replies. For any questions, please e-mail customerservice@vdxpathology.com

I-Stat Chem 8+ Serial/Recheck (In-House) (Ref: US798-DR436638)

Test	Results	Unit	Lowest Value	Highest Value	Qualifier
Na	147	mmol/L	139	150	
K	4.3	mmol/L	3.4	4.9	
Cl	106	mmol/L	106	127	
iCa	1.19	mmol/L	1.12	1.4	
TCO2	29	mmol/L	17	25	
Glu	100	mg/dL	60	115	
BUN	21	mg/dL	10	26	
Creat	1.1	mg/dL	0.5	1.3	
Hct	36	%	35	50	
Hb	102.2	g/dL	12.0	17.0	
Anion Gap	18	mmol/L	8	25	
Manual PCV	40	%	35	50	
Manual TS	5.6	g/dL	5.5	7.5	
Serum Color	Clear				

Test	Results	Unit	Lowest Value	Highest Value	Qualifier
Citrated Partial Thromboplastin Time (PTT)	121.0	seconds	72	102	
Citrated Prothrombin Time (PT)	14.0	seconds	11	17	

CT Scan (Ambra) (Ref: US798-DR436611)

Outcome**History:**

- History - 1 month history of exercise intolerance, decreased appetite and weight loss. Acute onset of dyspnea: presented to ER yesterday. Pleural effusion consistent with chylothorax. Echo showed hypocontractility, but no structural changes that could explain the chylothorax. The dog is severely lethargic/obtunded and ataxic, which is not typical of an idiopathic chylothorax.
- Lab Findings -
- Clinical Questions to be Addressed in Study -

FINDINGS:

A pre and post IV contrast CT of the abdomen dated 5/22/2026 is available for review. There is a moderate volume of contrast within the urinary bladder at the start of the study.

In the visible portion of the thorax, there is a moderate volume of right sided pleural effusion and mild volume of left-sided pleural effusion. There is mild contrast-enhancement of the parietal pleura bilaterally. There is a 6 mm x 7.8 mm ill-defined nodular region within the right middle lung lobe (series 14 image 13). There is mild wispy soft tissue attenuating regions within the subcutaneous tissues of the left lateral thorax. There is focal thickening of the mid body of the left sixth rib.

There is no evidence of peritoneal effusion or peritoneal gas. The gallbladder is moderately distended with ventrally dependent hyperattenuating material. The common bile duct is normal in size and can be traced to its termination at the duodenal papilla. There are multiple small hypoattenuating noncontrast enhancing hepatic nodules (all less than 5 mm in diameter). The spleen is normal with normal contrast enhancement. One of the splenic lymph nodes is mildly enlarged measuring 12 mm x 5.8 mm (series 12 image 226). The adrenal glands are normal. The kidneys are small with irregular margination bilaterally (left 4.3cm, right 4.2cm). There is renal diverticular mineralization bilaterally. There is nephrolithiasis bilaterally (Left: 6 mm x 6 mm x 17 mm, Right: 5.5 mm x 6.2 mm x 6.8 mm and 4.2 mm x 3.1 mm x 7 mm). Renal pelvis dilation is not identified. The ureters can be traced to their termination at the trigone. The urinary bladder contains a moderate amount of hyperattenuating contrast at the start of the study limiting evaluation of luminal contents. The visible urinary bladder wall is unremarkable. There are multiple regions of mineralization within the prostate. The prostate is normal in size measuring 1.7 cm x 1.4 cm. These regions of mineralization connect with the contrast column from within the urethra on the postcontrast series. The urethra is otherwise normal.

The pancreas is normal. The gastric lumen contains a moderate amount of gas and mild amount of fluid. The small intestines contain a mild amount of gas and fluid. The colon contains gas. One of the jejunal lymph nodes is mildly to moderately enlarged (11 mm x 11 mm, series 15 image 314).

The lumbar spinal column is well aligned. There is a mild to moderate amount of hyperattenuating material in the ventral to mildly left-sided spinal canal at the T13-L1 causing mild to moderate lifting and compression of the spinal cord at this site. There is mild hyperattenuating thickening of the dorsal annulus at L2-L3 compression of the spinal cord. There is moderate thickening of the femoral necks with osteophyte formation bilaterally. There is flattening and sclerosis of the acetabulum with osteophyte formation bilaterally.

CONCLUSION:

1. Mineralization within the prostate. Possible differentials include dystrophic mineralization, retained contrast within prostatic invaginations, or emerging neoplasia. BRAF testing could be considered.
2. Mild to moderate enlargement of one of the jejunal lymph nodes and mild enlargement of one of the splenic lymph nodes. Possible differentials include reactive or less likely metastatic lymphadenopathy.
3. Moderate bilateral renal degenerative changes with nephrolithiasis indicating chronic renal disease. Clinical significance be correlated with chemistry and urinalysis findings.
4. Multiple hypoattenuating hepatic nodules. Benign etiologies (e.g. nodular regeneration, hepatic cyst) are prioritized.
5. Moderate gallbladder distention with gravity-dependent hyperattenuating material. This likely represents cholecystolithiasis or mucinous hyperplasia.
6. Chronic intervertebral disc disease at T13-L1 moderate spinal cord compression. Chronic intervertebral disc disease at L2-L3 with minimal spinal cord compression. Clinical significance to be correlated to neurologic exam findings.

7. Bilateral pleural effusion with evidence of pleuritis/pleural fibrosis. This is consistent with the previously diagnosed chylothorax.
8. Ill-defined nodular region within the right middle lung lobe likely representing atelectasis as it is not identified on the previous days CT. A true pulmonary nodule is considered less likely.
9. Edema within the left lateral thoracic tissues. Possible differentials include hemorrhage/hematoma formation or leakage of pleural fluid into the subcutaneous tissues following thoracocentesis.
10. Hip dysplasia with moderate osteoarthritis of the coxofemoral joints bilaterally.
11. Healed fracture of the left sixth rib.

This report was generated using voice recognition software. As a result, minor typographical and grammatical errors may occur and potentially be confusing or misleading. Please contact me at dalal.hares@medvet.com if there are any concerns or questions about the content of this report.

This report was finalized by Dalal Hares, DVM, DACVR at 1:10 PM Mountain Time on 05/24/2026

Support Email: interviewsupport@vetrad.com

Support Phone: 888.483.8723

Clinic Notes / Specifics

[VIEW IMAGES](#)

IMAGE COUNT: 2457

External Link:

<https://medvet-cloud.ambrahealth.com/link/94234e37-d11f-4100-9f30-6e3a5f7a070b>

Chemistry(2) (Ref: US798-DR436188)

Test	Results	Unit	Lowest Value	Highest Value	Qualifier
Triglyceride	71	mg/dL	20	150	

Other(12) (Ref: US798-DR436188)

Test	Results	Unit	Lowest Value	Highest Value	Qualifier
Add-on Test Request 2					
Your Add On request has been processed.					

Endoscopic Images (Ambra) (Ref: US798-DR436365)

Clinic Notes / Specifics

[VIEW IMAGES](#)

IMAGE COUNT: 3

External Link:

<https://medvet-cloud.ambrahealth.com/link/2bd45966-59ec-4f44-91e3-1f612ad51f28>

CT Scan (Ambra) (Ref: US798-DR436321)

Outcome

History:

- History - History for BO: CT-scan thorax +/- add-on abdomen and head
1 month history of exercise intolerance, decreased appetite and weight loss. Acute onset of dyspnea: presented to ER yesterday. Pleural effusion consistent with chylothorax. Echo showed hypocontractility, but no structural changes that could explain the chylothorax. The dog is severely lethargic/obtunded and ataxic, which is not typical of an idiopathic chylothorax. Need lymphangiogram
- Lab Findings -
- Clinical Questions to be Addressed in Study -

FINDINGS:

A CT examination of the neck and thorax performed pre- and post- iodinated non-ionic contrast medium administration (Iohexol 240 mg/mL) on May 21st, 2026 is evaluated (3823 images). Transverse series in medium frequency reconstruction algorithm (soft tissue) and high frequency reconstruction algorithm (bone, lung) including sagittal and dorsal multiplanar reconstructions are provided for interpretation. A lymphangiogram was performed using a metatarsal pad approach. Both metatarsal pads were aseptically prepared. A 1.5 inch 22G butterfly catheter was inserted in each metatarsal pad. A volume of 1 ml/kg (9 ml total) of iodinated non-ionic contrast medium (Iohexol 240) divided in equal aliquots was injected in each metatarsal pad over 30 seconds. The metatarsal pads, distal hindlimbs and thighs were massaged for 4 minutes. The CT examination was performed 5 and 7 minutes after the metatarsal pad injections. Satisfactory lymphatic enhancement was obtained.

NECK:

In the ventral aspect of the larynx just caudal to the cuneiform tubercles, there is a ventral region of moderate and semi-circumferential soft tissue thickening that exhibits mild to moderate contrast-enhancement (pre mean: 51 HU, min: 20 HU, max: 71 HU; post mean: 83 HU, min: 56 HU, max: 125 HU) (approx. 0.99 cm thick x 2.2 cm wide). In the right ventral aspect of this soft tissue swelling, between the thyroid and cricoid cartilages, there is a curvilinear focus of mineral attenuation (0.37 cm). As a result, there is moderate narrowing of the laryngeal lumen. The hyoid apparatus is abnormal. The epiphyoid bones are decreased in size and non-mineralized. The ceratohyoid bones are mildly caudomedially displaced.

The palatine tonsils are normal and symmetrical in size. The oropharynx and nasopharynx are gas-filled. The mandibular and parotid salivary glands are normal in size, shape and attenuation. The medial retropharyngeal lymph nodes are mildly enlarged and rounded (right: 0.66 cm wide, left: 0.81 cm wide). The mandibular lymph nodes are normal and symmetrical in size. The deep cervical lymph nodes are not identified attribute to their very small size.

The thyroid glands are normal in size and attenuation. The external and middle ears are gas filled without mural thickening. There is minimal mineralization of the external ear canal cartilages bilaterally. The trachea and esophagus are normal. The cervical spine is normal. In the subcutaneous tissues along the right dorsolateral aspect of the neck at the level of C4 vertebra, there is a well-circumscribed, round and soft tissue attenuating nodule with a punctate mineralized center (0.46 cm).

THORAX:

In the pleural space bilaterally, there is a moderate amount of fluid that settles dependently in the thoracic cavity (mean: 23 HU, min: -5 HU, max: 51 HU). The pleural fluid separates the lung lobes and causes retraction of the ventral lung lobe margins from the thoracic wall. A minimal amount of gas is present in the dorsal aspect of the right pleural space. Best identified on the post-contrast series, there is mild to moderate thickening of the ventral mediastinal and costal parietal pleura. The lung lobe margins are rounded.

The ventral aspect of the right middle lung lobe and cranial subsegment of the left cranial lung lobe is severely hypoinflated such that the pulmonary parenchyma is effaced by regions of soft tissue attenuation. Multifocally throughout the remainder of the pulmonary parenchyma, there are parenchymal bands of soft tissue attenuation and patchy regions of ground-glass attenuation. There are no soft tissue attenuating pulmonary nodules. The bronchial tree is patent and normal in diameter. There is no evidence of bronchial thickening or dilation.

The pulmonary vasculature is normal without filling defects on the post-contrast series. The cardiac chambers are normal in size and wall thickness. The venous vasculature and major thoracic arteries are of normal tomography. The cranial mediastinal lymph nodes are mildly enlarged, and the surrounding mediastinal fat contains strands of fluid attenuation (cranial mediastinal lymph nodes up to 1.1 cm in diameter). The sternal tracheobronchial and axillary lymph nodes are normal in size, shape and attenuation. In the cranial aspect of the thoracic cavity, the esophagus is mildly distended with fluid.

The right kidney is moderately to severely reduced in size and markedly irregularly marginated. A moderate amount of amorphous mineralization is present in the diverticula and pelvis of the right kidney. The gallbladder is moderately distended and contains a large amount of partially mineralized and gravity-dependent debris. Multifocally in the included portion of the liver, there are rounded cysts filled with fluid. The stomach is moderately distended with fluid and gas. The remainder of the included abdomen is normal. In the subcutaneous tissues along the left ventral aspect of the thorax, there are numerous fluid strands that interdigitate with the subcutaneous fat.

LYMPHANGIOGRAM:

The collimated cranial portion of the cisterna chyli is located in the retroperitoneal space along the right dorsal aspect of the aorta ventral to L1 vertebra, and consists of two small converging and diverging lymphatic vessels. The caudal extent of the cisterna chyli is not included in the study. The thoracic duct lies retroperitoneally along the cranial abdominal aorta between the crura of the diaphragm and forms the cranial continuation of the cisterna chyli. There are two thoracic duct branches. Along T11, T12 and T13, the thoracic duct branches are respectively located along the right and left dorsal aspect of the aorta. Both branches converge, diverge, and are joined by short intermediate branches dorsal to the aorta. Cranial to T10 vertebra, both branches course separately along the descending aorta. Along the dorsal aspect of the left subclavian artery, both thoracic duct branches unite into a single thoracic duct that subsequently joins the cranial mediastinal lymph nodes and insert at the junction of the left external jugular veins with the left subclavian veins. Following the lymphangiogram study, there is a mild increase in the attenuation of the pleural effusion (pre IV contrast mean: 16 HU; post IV contrast mean: 17 HU, post lymphangiogram mean: 37 HU).

CONCLUSION:**NECK:**

1. Ventral laryngeal mass with central focus of mineral attenuation, causing partially attenuation of the laryngeal lumen. Differential diagnoses include granulation tissue secondary to prior surgical devocalization or foreign body migration (given the central focus of mineral attenuation) vs. neoplasia (e.g., squamous cell carcinoma, mast cell tumor, lymphoma, rhabdomyosarcoma, chondrosarcoma, oncocytoma, etc.).
2. Mild bilateral medial retropharyngeal lymphadenopathy. Reactive lymphoid hyperplasia is suspected. Metastatic neoplasia is a secondary consideration.
3. Bilateral hypoplasia of the epihyoid bone.

THORAX:

1. Moderate bilateral pleural effusion and mild to moderate pleuritis/pleural fibrosis, consistent with the recently diagnosed chylothorax. The minimal right sided pneumothorax is consistent with an iatrogenic change secondary to prior thoracocentesis.
2. Secondary mild to moderate multilobar pulmonary atelectasis.
3. Mild cranial mediastinal lymphadenopathy with perinodal effusion and/or steatitis, most consistent with underlying reactive lymphoid hyperplasia.
4. Mild esophageal fluid distension, likely secondary to gastroesophageal reflux under general anesthesia.
5. Moderate to severe right-sided chronic renal degeneration including nephrolithiasis.
6. Moderate gallbladder distention (likely secondary to fast), cholecystolithiasis and multifocal hepatic cysts vs. biliary cystadenomas.
7. Moderate gastric distention with fluid and gas, suggestive of gastric stasis (possibly induced by general anesthesia).
8. Mild subcutaneous fluid along the left ventral aspect of the thorax, most consistent with leakage of pleural fluid into the subcutaneous tissues at the time of prior thoracocentesis.

LYMPHANGIOGRAM:

1. Pleural lymphatic leakage as supported by the increased attenuation of the pleural effusion following the lymphangiogram study.
2. Two thoracic duct branches (right dorsal and left dorsal branches - see description above).
3. No evidence of lymphangiectasia.

This report was finalized by Alexandra Beaulieu, DVM, DVSc, DACVR at 2:10 PM Eastern Time on 05/22/2026

Support Email: interviewsupport@vetrad.com

Support Phone: 888.483.8723

Clinic Notes / Specifics[VIEW_IMAGES](#)

IMAGE COUNT: 3823

External Link:

<https://medvet-cloud.ambrahealth.com/link/07af4fe6-4b44-49b2-a64e-d1a0c001d87b>

(Ref: US798-DR436186)

Outcome

Attached is VDX Report #: 26052396

Owner: Nguyen, Thao, Pet: Bo US798-DR436186

Doctor: Kelly, Acct: 7070

This report has been sent from an automated system which does not accept replies. For any questions, please e-mail customerservice@vdxpathology.com

Echocardiogram In-House Initial (Ambra) (Ref: US798-DR436312)

Clinic Notes / Specifics[VIEW_IMAGES](#)

IMAGE COUNT: 48

External Link:

<https://medvet-cloud.ambrahealth.com/link/e36bdec2-ba0c-46a5-98bb-b7377e1c1f32>

Point of Care Ultrasound (Ambra) (Ref: US798-DR436164)

Outcome

moderate pleural effusion. subjectively decreased cardiac contractility. no pericardial effusion.

(Ref: US798-DR436190)

Outcome

see attachments

Snap 4DX+ Heartworm (IVLS) (Ref: US798-DR436194)

Outcome

negative for all

Radiographs Per Region (Ambra) (Ref: US798-DR436187)

Outcome

History:

- History - Progressive difficulty breathing over the past 3 days. Shallow breaths on exam. thoracocentesis- aspirated 68ml left and 375ml right side (suspected) chylous pleural effusion. rads are post-thoracocentesis
- Lab Findings - cbc/chem/lytes nsf. HW test negative
- Clinical Questions to be Addressed in Study - cause for chylous effusion

FINDINGS:

Three-view thoracic radiographs (3 images) performed on May 20th, 2026 are evaluated.

The pleural space of both hemithoraces contains a mild amount of fluid forming thick interlobar fissures that extend to the pulmonary hilus and separate the lung lobes. There is mild retraction of the lung lobes from the thoracic wall.

The right cranial lung lobe is severely hypoinflated and increased in soft tissue opacity. This lung lobe contains sharp and irregularly marginated air bronchograms. There is mild hypoinflation of the right middle lung lobe and both subsegments of the left cranial lung lobe. The remainder of the pulmonary parenchyma is normal and adequately inflated. The lung lobe margins are mildly rounded. The cardiovascular structures and mediastinum are normal. The trachea and principal bronchi are patent and normal in diameter.

The stomach is mildly distended with gas. Amorphous mineralization is present in the region of both kidneys. There is mild and smoothly marginated enlargement of the mid aspect of the left 6th rib. Minimal spondylosis deformans is present along the ventral aspect of T3-4 intervertebral disc space. A wet hair artifact is present along the thorax.

CONCLUSION:

1. Mild bilateral pleural effusion and pleuritis/restrictive pleuritis, consistent with the reported chylothorax.
2. Secondary atelectasis the right cranial, right middle and left cranial lung lobes.
3. Mild gastric gas distention, likely secondary to aerophagia.
4. Bilateral nephrolithiasis.
5. Chronic healed fracture of the left 6th rib.

An internal medicine consultation and CT lymphangiogram could be considered for further evaluation of the chylothorax.

This report was finalized by Alexandra Beaulieu, DVM, DVSc, DACVR at 12:08 PM Eastern Time on 05/21/2026
Support Email: interviewsupport@vetrad.com
Support Phone: 888.483.8723

Clinic Notes / Specifics

[VIEW IMAGES](#)

IMAGE COUNT: 3

External Link:

<https://medvet-cloud.ambrahealth.com/link/2c2fb93a-211b-497e-aaa9-113c976614a5>

Assessments

Problems:

- Chylous pleural effusion: r/o idiopathic/breed related vs cardiac vs lung lobe torsion vs infection vs neoplasia vs other
- Arrhythmia, subjectively decreased cardiac contractility on POCUS: r/o inflammation secondary to chylous effusion vs primary cardiac disease vs other
- Short shallow breaths progressive over 3 days: d/t chylous effusion

Hx:

- Glaucoma OU
- Debarked

Cli comm:

critical approved on arrival

spoke with owners about chylothorax, r/o's as above, also concerned that his heart may not be contracting normally. discussed hosp with chest tap, fluid out for analysis, chest rads, abd u/s, echo tomorrow. start with med mgmt, however if not working and chest re-fills quickly or continues to refill despite meds then CT and surgery are the next steps. CT and surgery (and associated costs at that time) w/o treatments and dx today can be 20-30k. We can see how he responds to meds but prognosis is not good, I do not see many that respond well to medical management, most that I have seen end up needing surgery and even that can have complications, not a guarantee fix.

presented est for initial dx and tx 1-2 days in hosp, approved. received pre-approval with trupanion

Dx:

- POCUS: moderate pleural effusion. hear has subjectively decreased contractility, arrhythmia, no pauses longer than 2 seconds
- Paired triglycerides to Idexx: pending
- Fluid analysis to VDX: pending
- 4Dx/HW test: negative for all
- CBC/chem 17/lytes: NSF

Plan:

IVC

O2 at 50%

Thoracocentesis

sedation: butorphanol 0.3mg/kg IV

local block lidocaine 0.5ml

have owners purchase Rutin

Norm-R 10ml/kg bolus, then 30ml/hr IV

own Rutin 500mg PO q8

own latanoprost 1 drop OU BID

own dorzolamide 1 drop OU q8

water always

bland low fat diet q8

no food after 12am in prep for u/s or other sedated diagnostics tomorrow

Shift summary:

breathing improved, stable after thoracocentesis despite there still being pleural effusion based on rads.
not interested in eating

transfer to IM in the morning for ongoing care

Planned dx (but ask DVM first):

echo

abd u/s

Daily Assessments

05-22-2026 - IM

S/O

Diagnostics/medical record review:

5/20/2026 - MV SV ER

- POCUS: moderate pleural effusion; heart has subjectively decreased contractility, arrhythmia, no pauses longer than 2 seconds.
- Thoracocentesis: Removal of 375 mL on R side, 68 mL on L side; fluid has a chylous appearance
- Paired triglycerides (idexx): **pending**
- Fluid analysis (VDx): Chylothorax - TNCC 800/uL, RBC <10,000/uL, Protein 4.7 g/dL
- CBC (IH): Hct 49.8%, no stress leuk
- Chem (IH): Crea 1.3, Cl 107 (L), Alb 3.0
- Snap 4dx: Neg x4
- CXR:
 - Mild bilateral pleural effusion and pleuritis/restrictive pleuritis, consistent with the reported chylothorax.
 - Secondary atelectasis the right cranial, right middle and left cranial lung lobes.
 - Mild gastric gas distention, likely secondary to aerophagia.
 - Bilateral nephrolithiasis.
 - Chronic healed fracture of the left 6th rib.

5/21/2026 - MV SV Hospit (IM)

- T-Fast: Moderate amount of pleural effusion bilaterally, R>L
- Neuro evaluation: Suspect that current status is due to extra-cranial cause(s), but cannot rule out a primary brain disease.
- Cardio consult with echocardiogram:
 - Structurally normal heart with decreased contractility - r/o extra-cardiac disease causing myocardial depression (infection, inflammation, neoplasia??) vs myocardial infarction (less likely given age and breed)
 - Mild to moderate pleural effusion with normal hepatic vessels - r/o extra-cardiac
 - Consider Pimobendan + recheck in 4-6 months
- General anesthesia for thoracocentesis followed by CT-scan - *see assessment*
- Thoracocentesis: removal of 180 mL of chylous fluid from R side without complications; no fluid seen on L side
- CT-scan (thorax + neck, with lymphangiogram):
 - NECK:
 - Ventral laryngeal mass with central focus of mineral attenuation, causing partial attenuation of the laryngeal lumen. Differential diagnoses include granulation tissue secondary to prior surgical devocalization or foreign body migration (given the central focus of mineral attenuation) vs. neoplasia (e.g., squamous cell carcinoma, mast cell tumor, lymphoma, rhabdomyosarcoma, chondrosarcoma, oncocytoma, etc.).
 - Mild bilateral medial retropharyngeal lymphadenopathy. Reactive lymphoid hyperplasia is suspected. Metastatic neoplasia is a secondary consideration.
 - Bilateral hypoplasia of the epiphyoid bone.
 - THORAX:
 - Moderate bilateral pleural effusion and mild to moderate pleuritis/pleural fibrosis, consistent with the recently diagnosed chylothorax. The minimal right sided pneumothorax is consistent with an iatrogenic change secondary to prior thoracocentesis.

- Secondary mild to moderate multilobar pulmonary atelectasis.
- Mild cranial mediastinal lymphadenopathy with perinodal effusion and/or steatitis, most consistent with underlying reactive lymphoid hyperplasia.
- Mild esophageal fluid distension, likely secondary to gastroesophageal reflux under general anesthesia.
- Moderate to severe right-sided chronic renal degeneration including nephrolithiasis.
- Moderate gallbladder distention (likely secondary to fast), cholecystolithiasis and multifocal hepatic cysts vs. biliary cystadenomas.
- Moderate gastric distention with fluid and gas, suggestive of gastric stasis (possibly induced by general anesthesia).
- Mild subcutaneous fluid along the left ventral aspect of the thorax, most consistent with leakage of pleural fluid into the subcutaneous tissues at the time of prior thoracocentesis.
- LYMPHANGIOGRAM:
 - Pleural lymphatic leakage as supported by the increased attenuation of the pleural effusion following the lymphangiogram study.
 - Two thoracic duct branches (right dorsal and left dorsal branches - see description above).
 - No evidence of lymphangiectasia.
- Biopsy of laryngeal mass: Pending
- Received triglyceride level: 71 -- we called idexx as we only received one result (supposed to be paired TG), and figured out they only ran it on the serum, so requested to run the same test on the chylous fluid (currently pending)

Current Treatments:

See SmartFlow for doses/routes/frequencies

- Norm-R
- Latanoprost
- Dorzolamide
- Rutin
- Pimobendan

Physical Exam

Weight: 10.2 kg

Temp (°F): 101.6 H.R.: 112 bpm R.R.: 24 rpm

C.R.T. :1-2 sec

M.M.: Pink, moist

Attitude: Very lethargic

Hydration: Euhydrated

B.C.S.: 6-7/9; MCS 3/3

P.S.: 2/4

CARDIOVASCULAR: Normal rhythm detected; no murmur ausculted; femoral pulses strong and synchronous.

RESPIRATORY: Eupneic, no respiratory effort; decreased bronchovesicular ventrally; inspiratory + expiratory stridor on auscultation.

ABDOMEN: Abdomen is very tensed and suspect painful on palpation; no overt organomegaly.

A

Problems/Diagnoses:

- Chylothorax - suspect idiopathic (normal thorax [CT-scan], normal lymphangiogram)
- Laryngeal mass - cancerous (carcinoma, lymphoma, other) vs. benign (inflammatory, granuloma) - histopathology pending
- Cardiac hypomotility - unclear etiology, suspect secondary to extra-cardiac disease
- Severe lethargy, dullness - unclear etiology: secondary to current respiratory condition vs. primary brain disease
- Suspect abdominal pain
- Hx of mature cataract and glaucoma OU
- Hx of "debark surgery" (prior to adoption by current O)

P

Diagnostic performed:

- T-Fast: Moderate amount of pleural effusion, R>L
- Thoracocentesis: removal of 300 mL of chylous fluid from R side without complications

- CT-scan abdomen: *It was planned to do an AUS, but we changed it for an additional site of CT - see assessment*
 - Mineralization within the prostate. Possible differentials include dystrophic mineralization, retained contrast within prostatic invaginations, or emerging neoplasia. BRAF testing could be considered.
 - Mild to moderate enlargement of one of the jejunal lymph nodes and mild enlargement of one of the splenic lymph nodes. Possible differentials include reactive or less likely metastatic lymphadenopathy.
 - Moderate bilateral renal degenerative changes with nephrolithiasis indicating chronic renal disease. Clinical significance be correlated with chemistry and urinalysis findings.
 - Multiple hypoattenuating hepatic nodules. Benign etiologies (e.g. nodular regeneration, hepatic cyst) are prioritized.
 - Moderate gallbladder distention with gravity-dependent hyperattenuating material. This likely represents cholecystolithiasis or mucinous hyperplasia.
 - Chronic intervertebral disc disease at T13-L1 moderate spinal cord compression. Chronic intervertebral disc disease at L2-L3 with minimal spinal cord compression. Clinical significance to be correlated to neurologic exam findings.
 - Bilateral pleural effusion with evidence of pleuritis/pleural fibrosis. This is consistent with the previously diagnosed chylothorax.
 - Ill-defined nodular region within the right middle lung lobe likely representing atelectasis as it is not identified on the previous days CT. A true pulmonary nodule is considered less likely.
 - Edema within the left lateral thoracic tissues. Possible differentials include hemorrhage/hematoma formation or leakage of pleural fluid into the subcutaneous tissues following thoracocentesis.
 - Hip dysplasia with moderate osteoarthritis of the coxofemoral joints bilaterally.
 - Healed fracture of the left sixth rib.
- iStat: TCO2 29 (H), Crea 1.1, PCV 40%, TS 5.6, serum clear
- PT/PTT: 14 (N) / 121 (HN)

Assessment/SWO

AM Communication:

Bo is fairly stable compared to yesterday: he is still very dull, and reacts heavily when we touch his abdomen. His breathing is still good (not dyspneic), but he has accumulated a significant amount of fluid around his lungs again. Reviewed with the owners the results from yesterday, and the concerns that remain with Bo. Discussed that we could retap him today and monitor, or consider sending him home (with the risks of worsening or even dying at home), or place chest tube to prevent several thoracocentesis throughout the weekend. The O do not want to move forward with chest tubes. They are considering taking him home, but are very concerned about him. Discussed imaging his abdomen with an US to make sure we are not missing something - O agree.

As we were about to do an AUS, Bo became dyspneic as soon as he was put on his back. He was stabilized by putting him sternal and administering flow-by O2. I repeat a thoracocentesis, and removed 300 mL of chylous fluid. Then I discussed with the radiology team and elected to do a CT-scan of the abdomen instead of an ultrasound: he is very dull and therefore immobile, and a CT will prevent putting him on his back and worsening his respiratory function. I also do not think that he will tolerate being pushed on the abdomen very much. As we did a CT-scan yesterday, we could charge for an "add-on site" and it was relatively similar in cost.

There was a large bruise on his ventral thorax where he received the echo. I did a PT/PTT to rule out a weird coagulopathy.

PM Communication:

The CT-scan of his abdomen revealed some chronic anomalies, some that could explain some painful reaction (disc, severe hip orthopedic disease), however nothing that explains why he is so dull. A primary neurological disease is not ruled out (would need an MRI). The O came to visit and we discussed a long time about Bo's prognosis, which is guarded to poor. Even if the laryngeal mass is benign, Bo still needs surgery in the near future to correct his chylothorax, and there is no guarantee that it will be successful. Discussed about euthanasia which is a fair option for Bo at this time. At the end of the discussion, the O elected to take Bo home - they understand that I cannot predict how long he will remain able to breathe well (he may need to come back to be tapped in 24 or 48 hours). They are considering humane euthanasia, but need to think about it. They took home some cards for at-home euthanasia services.

Updated problem list

- Chylothorax - suspect idiopathic (normal thorax [CT-scan], normal lymphangiogram)
- Laryngeal mass - cancerous (carcinoma, lymphoma, other) vs. benign (inflammatory, granuloma - from "debarb" surgery?) - histopathology pending
- Cardiac hypomotility - unclear etiology, suspect secondary to extra-cardiac disease
- Severe lethargy, dullness - unclear etiology: secondary to current respiratory condition vs. primary brain disease
- Suspect pain: back, hip, other?
- Hip dysplasia with moderate osteoarthritis of the coxofemoral joints bilaterally
- Moderate bilateral renal degenerative changes with nephrolithiasis - no azotemia

- Potential mineralization within the prostate vs. retained contrast within prostatic invaginations
- Healed fracture of the left sixth rib
- Hx of mature cataract and glaucoma OU
- Hx of "debark surgery" (prior to adoption by current O)

Treatments

CONTINUE:

- Rutin 500mg PO q8h
- Latanoprost OU q12h
- Dorzolamide OU q8h
- Pimobendan 2.5mg PO q12h

CHANGE:

- IVF: Stop

Plan for tonight/tomorrow:

- Discharge home tonight - the owners want to spend some time with him before potentially considering euthanasia

05-21-2026 - Overnight ER

S/O

History:

- Increased respiratory effort that started on Sunday (4 days ago), progressively worse yesterday which triggered the ER visit.
- For about 3-4 weeks, he has been getting more tired recently, not able to walk as far as before. His appetite has also gone down, he will sometimes skip meals. He seems to have lost weight.
- Hx of mature cataracts + glaucoma: on Latanoprost + Dorzolamide OU
- Mostly outdoor dog
- UTD on vaccines (dapp, rabies, bordetella); receives simparica trio
- Adopted by current O in 2021: was already adult at that time, but age is unknown. Was already "debarked".

Physical Exam

Weight: 9.6 kg

Temp (°F): 99.4 H.R.: 80 bpm R.R.: 18 rpm

C.R.T.: 1-2 sec

M.M.: Pink, moist

Attitude: sedate

Hydration: euhydrated

B.C.S.: 6-7/9; MCS 3/3

P.S.: 0/24

EENT: Eyes – moderate mucoid discharge and irregular surface of cornea, mature cataracts OU; impossible to evaluate fundus due to cataracts. Nose – no discharge, no discoloration. Face – facial expression symmetrical. Teeth/mouth – moderate tartar. Ears – clean, no odor or discharge.

LYMPH NODES: Peripheral lymph nodes palpate smooth, soft, and symmetric.

CARDIOVASCULAR: Normal rhythm detected; no murmur ausculted; femoral pulses strong and synchronous.

RESPIRATORY: Mild inspiratory effort with upper airway stridor; decreased bronchovesicular ventrally (almost diffusely); dry crackles ventrally; inspiratory stridor on auscultation.

INTEGUMENT: Hair coat is clean, shiny, and full; skin – no masses, lesions or erythema present.

ABDOMEN: Abdomen is soft and non-painful on palpation; no overt organomegaly.

MUSCULOSKELETAL: Ambulatory when strongly motivated to get up - generalized weakness and mild ataxia; no joint effusion or pain noted; normal muscle mass.

NEUROLOGIC: Very lethargic, almost obtunded; cranial nerves intact (except PLR) and normally elicited; normal paw placement.

UROGENITAL: External genitalia appear within normal limits.

RECTAL: Normal – no masses, anal sacs empty, normal to soft stools on glove.

A

Assessment/SWO

Updated problem list

- Chylothorax - idiopathic
- Laryngeal mass - cancerous (carcinoma, lymphoma, other) vs. benign (inflammatory, granuloma) - histopathology pending
- Cardiac hypomotility - unclear etiology
- Severe lethargy and ataxia - secondary to respiratory disease vs. concurrent intra-cranial involvement (thromboembolism, neoplasia, infectious)
- Hx of mature cataract and glaucoma OU

P

Diagnostic performed:

- None overnight

Treatments

CONTINUE:

- Rutin 500mg PO q8h
- Latanoprost OU q12h
- Dorzolamide OU q8h
- IVF: Norm-R decreased to 20 mL/hr
- Pimobendan 2.5mg PO q12h per Dr. Chan's recommendations
- Dexamethasone 0.1mg/kg IV 1x after biopsy of laryngeal mass

ADDED:

- butorphanol 4 mg IV PRN

06:00 - P had an episode of distress, responded well to butorphanol and flow-by O2. Tapered off.

Communication

Good morning, it's Dr. Tupper from MEdVet. Bo unfortunately had one episode where he got really worked up and needed some meds to help him relax and catch his breath, but otherwise did okay though the night. Dr. Brassard will be in touch this morning to discuss his plan of care

Plan for tonight/tomorrow:

- Monitor Bo's respiration closely overnight: monitor RR, effort, evidence of stridor.
 - Keep calm, avoid overtly stressing the patient. Bo has been very lethargic, therefore anti-anxiety meds (trazodone, gabapentin, acepromazine, etc.) were not needed - however they can be considered if Bo becomes more alert and overtly anxious/agitated.
- If evidence of upper airway obstruction: intubate with a SMALL endotracheal tube: size 4.0 or smaller VS. consider thoracotomy
- Repeat T-Fast tomorrow AM

Jacob Tupper, DVM

05-21-2026 - Transfer to IM

S/O

History:

- Increased respiratory effort that started on Sunday (4 days ago), progressively worse yesterday which triggered the ER visit.
- For about 3-4 weeks, he has been getting more tired recently, not able to walk as far as before. His appetite has also gone down, he will sometimes skip meals. He seems to have lost weight.
- Hx of mature cataracts + glaucoma: on Latanoprost + Dorzolamide OU
- Mostly outdoor dog
- UTD on vaccines (dapp, rabies, bordetella); receives simparica trio
- Adopted by current O in 2021: was already adult at that time, but age is unknown. Was already "debarked".

Diagnostics/medical record review:

5/20/2026 - MV SV ER

- POCUS: moderate pleural effusion; heart has subjectively decreased contractility, arrhythmia, no pauses longer than 2 seconds.
- Thoracentesis: Removal of 375 mL on R side, 68 mL on L side; fluid has a chylous appearance
- Paired triglycerides (idexx): **pending**
- Fluid analysis (VDx): Chylothorax - TNCC 800/uL, RBC <10,000/uL, Protein 4.7 g/dL
- CBC (IH): Hct 49.8%, no stress leuk
- Chem (IH): Crea 1.3, Cl 107 (L), Alb 3.0
- Snap 4dx: Neg x4
- CXR:
 - Mild bilateral pleural effusion and pleuritis/restrictive pleuritis, consistent with the reported chylothorax.
 - Secondary atelectasis the right cranial, right middle and left cranial lung lobes.
 - Mild gastric gas distention, likely secondary to aerophagia.
 - Bilateral nephrolithiasis.
 - Chronic healed fracture of the left 6th rib.

Current Treatments:

See SmartFlow for doses/routes/frequencies

- Norm-R
- Latanoprost
- Dorzolamide
- Rutin

Physical Exam

Weight: 9.6 kg

Temp (°F): 100.50 H.R.: 100 bpm R.R.: 30 rpm

C.R.T. :1-2 sec

M.M.: Pink, moist

Attitude: Very lethargic

Hydration: <5% dehydrated

B.C.S.: 6-7/9; MCS 3/3

P.S.: 0/24

EENT: Eyes – moderate mucoid discharge and irregular surface of cornea, mature cataracts OU; impossible to evaluate fundus due to cataracts. Nose – no discharge, no discoloration. Face – facial expression symmetrical. Teeth/mouth – moderate tartar. Ears – clean, no odor or discharge.

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UROGENITAL: External genitalia appear within normal limits.

RECTAL: Normal – no masses, anal sacs empty, normal to soft stools on glove.

A**Problems/Diagnoses:**

- Chylothorax - thoracic duct obstruction (neoplasia, granuloma, abscess, other) vs. thrombus vs. CHF vs. lung lobe torsion vs. idiopathic
- Severe lethargy and ataxia - secondary to respiratory disease vs. concurrent intra-cranial involvement (thromboembolism, neoplasia, infectious)
- Hx of mature cataract and glaucoma OU

P**Diagnostic performed:**

- T-Fast: Moderate amount of pleural effusion bilaterally, R>L

- Neuro evaluation: Suspect that current status is due to extra-cranial cause(s), but cannot rule out a primary brain disease.
- Cardio consult with echocardiogram:
 - Structurally normal heart with decreased contractility - r/o extra-cardiac disease causing myocardial depression (infection, inflammation, neoplasia??) vs myocardial infarction (less likely given age and breed)
 - Mild to moderate pleural effusion with normal hepatic vessels - r/o extra-cardiac
 - Consider Pimobendan + recheck in 4-6 months
- General anesthesia for thoracocentesis followed by CT-scan - *see assessment*
- Thoracocentesis: removal of 180 mL of chylous fluid from R side without complications; no fluid seen on L side
- CT-scan (thorax + neck, with lymphangiogram): *Final report pending*
- Biopsy of laryngeal mass: Pending
- Received triglyceride level: 71 -- we called idexx as we only received one result (supposed to be paired TG), and figured out they only ran it on the serum, so requested to run the same test on the chylous fluid (currently pending)

Assessment/SWO

AM Communication:

Discussed with O about chylothorax including pathophysiology and causes. Cardiac origin was ruled out, however the cardiologist noticed hypercontractility, which can occur secondary to a systemic issue. Along with his severe lethargy, it worries me about an aggressive cause of chylothorax - it is indeed not typical to see them like this with IC due to the chronicity of the condition. However it remains a possibility as Shibas are overrepresented with IC. Discussed that next step is to perform a CT-scan of his thorax - and based on result, we may have to extend to abdomen and even head. Went over the risks of anesthesia with a dog that is sick like Bo. O agrees with the plan.

General anesthesia assessment:

My technician was not able to intubate even with very small size tubes due to some obstruction, and there was a significant amount of bleeding in the laryngeal area. I took over and successfully intubated with a ET size 4.0, significantly smaller than the size of his trachea. There was some structural obstruction that could be felt right caudally to the arytenoid, and I suspected a mass. The priority was to secure the airway which was done with intubation and suction of the blood. Bo was then stable with good SpO2 reading, therefore we moved forward with thoracocentesis and CT-scan - we included the neck to the area. The scan revealed a soft tissue opacity in the larynx consistent with the mass I suspected. However there was no intra-thoracic anomalies that could explain the chylothorax. We moved forward with the lymphangiogram, which was mostly unremarkable. This is very interesting as it suggests two different disease processes: laryngeal mass and idiopathic chylothorax. Before waking up, I evaluated the laryngeal and rostral aspect of trachea with the bronchoscope (no charge), which showed an irregular and friable soft tissue mass that was bleeding significantly. I obtained a biopsy which was submitted for histopathology. Bo then woke up slowly and uneventfully from GA. An anti-inflammatory dose of dexamethasone SP was given to prevent swelling and airway obstruction. No evidence of obstruction during post-procedure recovery.

PM Communication:

Discussed with the O about the results of the procedure. As we are still missing some pieces of the puzzle, we are most likely dealing with two unrelated disease processes with Bo. I suspect that the laryngeal mass was therefore before/not overtly symptomatic, and that it was discovered because of the current chylothorax. The laryngeal mass however does make sense of his increased inspiratory effort and inspiratory stridor (under auscultation only). Discussed that treatment will mostly depend on the etiology of this mass. Indeed for his chylothorax, we are talking about intra-thoracic surgery to ligate thoracic duct +/- pericardectomy - as IC in Shibas is associated with a fair prognosis with this procedure, it is a very invasive surgery that I would not recommend if his laryngeal mass is consistent with a bad cancer (it does not seem fair to put him through that if he unfortunately has a poor prognosis no matter what). O agrees. If it is a benign etiology (inflammatory growth, granuloma, other), then we could consider debulking/removal as well as surgical treatment for the IC. Even if benign however, Bo will need most likely need some pretty intense surgical procedures without a guaranty of success, and therefore his prognosis remains very guarded. The O want to do what is best for him, and are willing to consider surgical treatment in the future if indicated.

At this point, the plan is to keep Bo as stable as possible until we receive the result of the biopsy. I did tap his chest today and removed the majority of the fluid - we will keep him overnight and monitor him closely. I will recheck his pleural effusion tomorrow to assess how fast it is reaccumulating. He may need to be tapped again. O understand. If he is stable without major re-accumulation, he could be discharge tomorrow.

Updated problem list

- Chylothorax - idiopathic
- Laryngeal mass - cancerous (carcinoma, lymphoma, other) vs. benign (inflammatory, granuloma) - histopathology pending
- Cardiac hypomotility - unclear etiology
- Severe lethargy and ataxia - secondary to respiratory disease vs. concurrent intra-cranial involvement (thromboembolism, neoplasia, infectious)

- Hx of mature cataract and glaucoma OU

Treatments

CONTINUE:

- Rutin 500mg PO q8h
- Latanoprost OU q12h
- Dorzolamide OU q8h

CHANGE:

- IVF: Norm-R decreased to 20 mL/hr following GA

ADD:

- Pimobendan 2.5mg PO q12h per Dr. Chan's recommendations
- Dexamethasone 0.1mg/kg IV 1x after biopsy of laryngeal mass

Plan for tonight/tomorrow:

- Monitor Bo's respiration closely overnight: monitor RR, effort, evidence of stridor.
 - Keep calm, avoid overtly stressing the patient. Bo has been very lethargic, therefore anti-anxiety meds (trazodone, gabapentin, acepromazine, etc.) were not needed - however they can be considered if Bo becomes more alert and overtly anxious/agitated.
- If evidence of upper airway obstruction: intubate with a SMALL endotracheal tube: size 4.0 or smaller VS. consider thoracotomy
- Repeat T-Fast tomorrow AM

05/20/2026

PM Assessment

Physical Exam

Weight: 10.2 kg
 Temp (°F): 99.90 H.R.: 60 bpm R.R.: 30 rpm
 C.R.T.: 1-2 sec
 M.M.: Pink
 Attitude: Sedate
 Hydration: <5% dehydrated
 B.C.S.: 6/9
 P.S.: 0/24

EENT:

Dentition: oral exam not performed due to temperament

Eyes: Sclera and conjunctiva unremarkable. Cornea and anterior chamber clear.

Ears: Clean. Nose: No nasal discharge. Normal nasal planum.

LYMPH NODES: The submandibular, prescapular, axillary, inguinal and popliteal lymph nodes are normal.

CARDIOVASCULAR: No murmur auscultated. No arrhythmia auscultated. Femoral pulses are strong and synchronous.

RESPIRATORY: Eupneic. Lung sounds slightly muffled, no crackles or wheezes auscultated.

INTEGUMENT: Good skin turgor. Clean, full coat.

ABDOMEN: Pliable with no reaction on palpation. No mass or organomegaly palpated. The bladder palpated normally.

MUSCULOSKELETAL: Not witnessed ambulating; MCS 3/3

NEUROLOGIC: Menace, palpebral, and physiologic nystagmus present OU. Face and pupils symmetric.

UROGENITAL: The external genitalia appear normal.

RECTAL: External normally

Diagnostics

- Pending fluid cytology
- Pending paired triglycerides
- AUS scheduled for today AM

Progress Notes

- Rested comfortably overnight
- Food held in preparation for AUS today
- Eupneic throughout the night

Client communication

- 7am text: Good morning this is Dr. Letsinger from MedVet! Bo rested comfortably and was breathing well overnight. We will discuss his case in rounds, and the Internal Medicine team will reach out to discuss the plan for the day. Thank you!

Assessment

- Chylous pleural effusion: r/o idiopathic/breed related vs cardiac vs lung lobe torsion vs infection vs neoplasia vs other
- Arrhythmia, subjectively decreased cardiac contractility on POCUS: r/o inflammation secondary to chylous effusion vs primary cardiac disease vs other
- Short shallow breaths progressive over 3 days: d/t chylous effusion
- Historic cataracts OU
- Historic glaucoma OU
- Debarked previously

Ongoing treatments

- Norm-R @ 30mL/hr IV
- Rutin 50mg/kg (500mg) PO q 8hrs
- Latanoprost 1 drop OU q 12hrs
- Dorzolamide 1 drop OU q 8hrs

Planned action list

- Fasted ON in preparation for AUS
- Transfer to IM in AM
- IH echo scheduled today

Chloe Letsinger, DVM

Medication

Date/Time	Drug Name	Qty	Instructions	Prescribed By
05-22-2026 5:12:09pm	Pimobendan 2.5mg (Per Tablet)	10	Give 1 tablet by mouth every 12 hours until recheck.	Camille Brassard
05-22-2026 5:12:09pm	Gabapentin 100mg (Per Capsule)	15	Give 1 capsule by mouth as needed to keep Bo calm. Give max every 8 hours.	Camille Brassard
05-22-2026 2:00:00pm	Propofol 10mg/mL Inj (Per mL)	2.4		Camille Brassard
05-22-2026 1:00:00pm	Methadone 10mg/ml Inj (Per mL)	0.19		Camille Brassard
05-22-2026 11:00:00am	Pimobendan 2.5mg (Per Tablet)	1	Give #INPUT# tablet(s) by mouth #INPUT# time(s) a day.	Camille Brassard
05-22-2026 10:00:00am	Methadone 10mg/ml Inj (Per mL)	0.19		Camille Brassard
05-22-2026 5:00:00am	Butorphanol 10mg/mL Inj (Per mL)	0.4		Jacob Tupper
05-21-2026 11:00:00pm	Pimobendan 2.5mg (Per Tablet)	1	Give #INPUT# tablet(s) by mouth #INPUT# time(s) a day.	Jacob Tupper

05-21-2026 3:20:01pm	Flumazenil 0.1mg/mL Inj (Per mL)	0.38		Camille Brassard
05-21-2026 2:50:01pm	Dexamethasone SP 4 mg/mL Inj (Per mL)	0.24		Camille Brassard
05-21-2026 1:26:38pm	Iohexol 240mg/mL Inj (Per mL)	33		Alexandra Beaulieu
05-21-2026 1:13:46pm	MIDazolam 5mg/mL Inj (Per mL)	0.38		Camille Brassard
05-21-2026 1:13:45pm	Butorphanol 10mg/mL Inj (Per mL)	0.2		Camille Brassard
05-21-2026 11:00:00am	Pimobendan 2.5mg (Per Tablet)	1	Give #INPUT# tablet(s) by mouth #INPUT# time(s) a day.	Camille Brassard
05-20-2026 6:00:00pm	Butorphanol 10mg/mL Inj (Per mL)	0.306		Robin Kelly
05-20-2026 4:00:15pm	Iohexol 240mg/mL Inj (Per mL)	26		Dalal Hares

Recommendations

SAIM DISCHARGE

Diagnosis/Differential:

- Chylothorax - suspect idiopathic (normal thorax [CT-scan], normal lymphangiogram)
- Laryngeal mass - cancerous vs. benign (inflammatory, granuloma) - histopathology pending
 - Causing partial upper airway obstruction
- Cardiac hypomotility - unclear etiology: suspect secondary to extra-cardiac disease
- Severe lethargy, dullness - unclear etiology: secondary to current respiratory condition vs. primary brain disease
- Severe hip osteoarthritis
- TL intervertebral disc disease, chronic
- Hx of mature cataract and glaucoma OU
- Hx of "debark surgery" (prior to adoption)

Bo is very sick, and his prognosis is considered guarded to poor unfortunately. At this time, even if his laryngeal mass is benign, he will eventually need invasive surgeries to correct his current respiratory diseases, with no guarantee of success. It is very fair to also consider humane euthanasia in this situation. Take some time at home to think about the options for Bo.

Recommendations

1) Medication(s):

- Pimobendan 2.5mg: Give 1 tablet by mouth every 12 hours until recheck.
 - Next dose: tonight around 10pm
- Rutin 500mg: Give 1 tablet by mouth every 8 hours until recheck.
 - Next dose: tonight around 8pm
- Gabapentin 100mg: Give 1 capsule by mouth as needed to keep Bo calm. Give max every 8 hours.
- Latanoprost and Dorzolamide: Continue to apply in both eyes as previously prescribed.

2) Diet:

- Bo can eat any diet he wants, though diets that are low in fat are ideal with his chylothorax.
- Due to his laryngeal mass that affects his swallowing ability, feed only wet or liquified food.

3) Recommendation(s):

- Bo has two respiratory diseases and is at high risk of respiratory distress. **Keep him calm** at all time to prevent dyspnea. Every episode of respiratory distress can unfortunately be life-threatening. As needed, give Bo the gabapentin prescribed.
- Monitor Bo's respiration very closely at home: if he starts to breathe heavily or rapidly, or if he seems to struggle to breathe, **bring him in to the emergency department IMMEDIATELY.**
- Bo's pleural effusion has been accumulating quickly while in hospital, and he will likely need to come to the hospital (ER department) to be tapped again this weekend.

Follow-Up

- I will contact you with the pending result of Bo's laryngeal mass biopsy.

Client Discharges

Diagnosis/Differential:

- Chylothorax - suspect idiopathic (normal thorax [CT-scan], normal lymphangiogram)
- Laryngeal mass - cancerous vs. benign (inflammatory, granuloma) - histopathology pending
 - Causing partial upper airway obstruction
- Cardiac hypomotility - unclear etiology: suspect secondary to extra-cardiac disease
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- Monitor Bo's respiration very closely at home: if he starts to breathe heavily or rapidly, or if he seems to struggle to breathe, **bring him in to the emergency department IMMEDIATELY.**
- Bo's pleural effusion has been accumulating quickly while in hospital, and he will likely need to come to the hospital (ER department) to be tapped again this weekend.

Follow-Up

- I will contact you with the pending result of Bo's laryngeal mass biopsy.

Thank you for giving us the opportunity to help in the care of Bo.

Camille Brassard, DVM, MS, DACVIM (SAIM)

Internal Medicine Department
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Fax: (408) 649-7072
Email: InternalMedicine.siliconvalley@medvet.com

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Patient Name: Nguyen^Bo	Patient ID: 645583	Institution: MedVet Silicon Valley
Patient Birth Date: 05/20/2018	Age: 8	Requester: Robin Kelly
Species: Canine	Sex: MN	Modality: DX
Breed: Shiba Inu	Accession: US798-DR436187	Study Date: 05/21/2026

Final Report

CASE DETAILS:

- History - Progressive difficulty breathing over the past 3 days. Shallow breaths on exam. thoracocentesis-aspirated 68ml left and 375ml right side (suspected) chylous pleural effusion. rads are post-thoracocentesis
- Lab Findings - cbc/chem/lytes nsf. HW test negative
- Clinical Questions to be Addressed in Study - cause for chylous effusion

FINDINGS:

Three-view thoracic radiographs (3 images) performed on May 20th, 2026 are evaluated.

The pleural space of both hemithoraces contains a mild amount of fluid forming thick interlobar fissures that extend to the pulmonary hilus and separate the lung lobes. There is mild retraction of the lung lobes from the thoracic wall.

The right cranial lung lobe is severely hypoinflated and increased in soft tissue opacity. This lung lobe contains sharp and irregularly marginated air bronchograms. There is mild hypoinflation of the right middle lung lobe and both subsegments of the left cranial lung lobe. The remainder of the pulmonary parenchyma is normal and adequately inflated. The lung lobe margins are mildly rounded. The cardiovascular structures and mediastinum are normal. The trachea and principal bronchi are patent and normal in diameter.

The stomach is mildly distended with gas. Amorphous mineralization is present in the region of both kidneys. There is mild and smoothly marginated enlargement of the mid aspect of the left 6th rib. Minimal spondylosis deformans is present along the ventral aspect of T3-4 intervertebral disc space. A wet hair artifact is present along the thorax.

CONCLUSION:

1. Mild bilateral pleural effusion and pleuritis/restrictive pleuritis, consistent with the reported chylothorax.
2. Secondary atelectasis the right cranial, right middle and left cranial lung lobes.
3. Mild gastric gas distention, likely secondary to aerophagia.
4. Bilateral nephrolithiasis.
5. Chronic healed fracture of the left 6th rib.

An internal medicine consultation and CT lymphangiogram could be considered for further evaluation of the chylothorax.

This report was finalized by Alexandra Beaulieu, DVM, DVSc, DACVR at 12:08 PM Eastern Time on 05/21/2026

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Acct # 7070

MedVet Silicon Valley**7090 Santa Teresa Blvd.****San Jose, CA 95139****Report No.****26052396****Phone: (408) 649-7070**Name **Bo US798-DR436186**Owner **Nguyen, Thao**Doctor **Kelly**Sp, Breed **K9, Shiba Inu**Sex **MN**DOB/Age **05/20/2018**Collected **5/20/26**Received **5/21/26**Reported **5/21/26**

HISTORY

Progressive difficulty breathing over the past three days. Ocular meds for glaucoma, no other known problems.

SPECIMEN

Pleural effusion: 5 mL of milky, cream colored, fluid is received in one WTT.

TNCC: 800/uL

RBC: <10,000/uL

Protein: 4.7 g/dL

MICROSCOPIC DESCRIPTION

Direct (A, B), sediment (C, D), and cytocentrifuge (E, F) smears are prepared from the submitted fluid and evaluated. The sample has a dense pale blue background with numerous fine free lipid vacuoles and scattered erythrocytes. In a 100 leukocyte differential, performed on the sediment smears, small lymphocytes predominate (86%) admixed with occasional larger lymphocytes and plasma cells, non-degenerate neutrophils (12%), few mildly vacuolated macrophages (1%) and eosinophils (1%). Greater numbers of macrophages are noted in the cytocentrifuge smears and they are sometimes appreciated in larger aggregates in the sediment smears. Few reactive mesothelial cells and rare individual mast cells are noted in the cytocentrifuge smears.

DIAGNOSIS

Chylous effusion

COMMENT

There is no overt evidence of exfoliating neoplasia in this sample. Chyle accumulates in body cavities when there is disruption of the lymphatic flow due to trauma, mass lesions (neoplasia or inflammatory), organ torsion, cardiovascular disease, or idiopathic causes. Correlate with clinical and imaging impressions.

PATHOLOGIST

Diana Schwartz, DVM

Diplomate, American College of

Veterinary Pathologists

Veterinarians: For consultation on this report, questions or to request additional testing, please e-mail CustomerService@vdxpathology.com or call (530) 753-4285.

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Acct # **7070**
MedVet Silicon Valley
7090 Santa Teresa Blvd.
San Jose, CA 95139

Phone: (408) 649-7070

Report No.
26052396

Name **Bo US798-DR436186**

Owner **Nguyen, Thao**

Doctor **Kelly**

Sp, Breed **K9, Shiba Inu**

Sex **MN**

DOB/Age **05/20/2018**

Collected **5/20/26**

Received **5/21/26**

Reported **5/21/26**

Cytology - Body Fluid

 BO NGUYEN

PET OWNER:	NGUYEN	MedVet Silicon Valley	LAB ID:	7201141323
SPECIES:	Canine	7090 Santa Teresa Blvd,	ORDER ID:	323466999
BREED:	Shiba Inu	San Jose, CA 95139	COLLECTION DATE:	5/20/26
GENDER:	Male Neutered	408-649-7070	DATE OF RECEIPT:	5/21/26
AGE:	8 Years	ACCOUNT #: 7918	DATE OF RESULT:	5/21/26
PATIENT ID:	548087	ATTENDING VET: Robin Kelly		

IDEXX Services: **Triglycerides, Serum**

Chemistry

5/21/26 (Order Received)
5/21/26 3:46 PM (Last Updated)

TEST	RESULT	REFERENCE VALUE	
Triglyceride	71	20 - 150 mg/dL	

Patient Name: Nguyen^Bo	Patient ID: 645583	Institution: MedVet Silicon Valley
Patient Birth Date: 05/20/2018	Age: 8	Requester: Camille Brassard
Species: Canine	Sex: MN	Modality: CT
Breed: Shiba Inu	Accession: US798-DR436321	Study Date: 05/21/2026

Final Report

CASE DETAILS:

- History - History for BO: CT-scan thorax +/- add-on abdomen and head 1 month history of exercise intolerance, decreased appetite and weight loss. Acute onset of dyspnea: presented to ER yesterday. Pleural effusion consistent with chylothorax. Echo showed hypocontractility, but no structural changes that could explain the chylothorax. The dog is severely lethargic/obtunded and ataxic, which is not typical of an idiopathic chylothorax. Need lymphangiogram
- Lab Findings -
- Clinical Questions to be Addressed in Study -

FINDINGS:

A CT examination of the neck and thorax performed pre- and post- iodinated non-ionic contrast medium administration (Iohexol 240 mg/mL) on May 21st, 2026 is evaluated (3823 images). Transverse series in medium frequency reconstruction algorithm (soft tissue) and high frequency reconstruction algorithm (bone, lung) including sagittal and dorsal multiplanar reconstructions are provided for interpretation. A lymphangiogram was performed using a metatarsal pad approach. Both metatarsal pads were aseptically prepared. A 1.5 inch 22G butterfly catheter was inserted in each metatarsal pad. A volume of 1 ml/kg (9 ml total) of iodinated non-ionic contrast medium (Iohexol 240) divided in equal aliquots was injected in each metatarsal pad over 30 seconds. The metatarsal pads, distal hindlimbs and thighs were massaged for 4 minutes. The CT examination was performed 5 and 7 minutes after the metatarsal pad injections. Satisfactory lymphatic enhancement was obtained.

NECK:

In the ventral aspect of the larynx just caudal to the cuneiform tubercles, there is a ventral region of moderate and semi-circumferential soft tissue thickening that exhibits mild to moderate contrast-enhancement (pre mean: 51 HU, min: 20 HU, max: 71 HU; post mean: 83 HU, min: 56 HU, max: 125 HU) (approx. 0.99 cm thick x 2.2 cm wide). In the right ventral aspect of this soft tissue swelling, between the thyroid and cricoid cartilages, there is a curvilinear focus of mineral attenuation (0.37 cm). As a result, there is moderate narrowing of the laryngeal lumen. The hyoid apparatus is abnormal. The epiphyoid bones are decreased in size and non-mineralized. The ceratohyoid bones are mildly caudomedially displaced.

The palatine tonsils are normal and symmetrical in size. The oropharynx and nasopharynx are gas-filled. The mandibular and parotid salivary glands are normal in size, shape and attenuation. The medial retropharyngeal lymph nodes are mildly enlarged and rounded (right: 0.66 cm wide, left: 0.81 cm wide). The mandibular lymph nodes are normal and symmetrical in size. The deep cervical lymph nodes are not identified attribute to their very small size.

The thyroid glands are normal in size and attenuation. The external and middle ears are gas filled without mural thickening. There is minimal mineralization of the external ear canal cartilages bilaterally. The trachea and esophagus are normal. The cervical spine is normal. In the subcutaneous tissues along the right dorsolateral aspect of the neck at the level of C4 vertebra, there is a well-circumscribed, round and soft tissue attenuating nodule with a punctate mineralized center (0.46 cm).

THORAX:

In the pleural space bilaterally, there is a moderate amount of fluid that settles dependently in the thoracic cavity (mean: 23 HU, min: -5 HU, max: 51 HU). The pleural fluid separates the lung lobes and causes retraction of the ventral lung lobe margins from the thoracic wall. A minimal amount of gas is present in the dorsal aspect of the right pleural space. Best identified on the post-contrast series, there is mild to moderate thickening of the ventral mediastinal and costal parietal pleura. The lung lobe margins are rounded.

The ventral aspect of the right middle lung lobe and cranial subsegment of the left cranial lung lobe is severely hypoinflated such that the pulmonary parenchyma is effaced by regions of soft tissue attenuation. Multifocally throughout the remainder of the pulmonary parenchyma, there are parenchymal bands of soft tissue attenuation and patchy regions of ground-glass attenuation. There are no soft tissue attenuating pulmonary nodules. The bronchial tree is patent and normal in diameter. There is no evidence of bronchial thickening or dilation.

The pulmonary vasculature is normal without filling defects on the post-contrast series. The cardiac chambers are normal in size and wall thickness. The venous vasculature and major thoracic arteries are of normal tomography. The cranial mediastinal lymph nodes are mildly enlarged, and the surrounding mediastinal fat contains strands of fluid attenuation (cranial mediastinal lymph nodes up to 1.1 cm in diameter). The sternal tracheobronchial and axillary lymph nodes are normal in size, shape and attenuation. In the cranial aspect of the thoracic cavity, the esophagus is mildly distended with fluid.

The right kidney is moderately to severely reduced in size and markedly irregularly marginated. A moderate amount of amorphous mineralization is present in the diverticula and pelvis of the right kidney. The gallbladder is moderately distended and contains a large amount of partially mineralized and gravity-dependent debris. Multifocally in the included portion of the liver, there are rounded cysts filled with fluid. The stomach is moderately distended with fluid and gas. The remainder of the included abdomen is normal. In the subcutaneous tissues along the left ventral aspect of the thorax, there are numerous fluid strands that interdigitate with the subcutaneous fat.

LYMPHANGIOGRAM:

The collimated cranial portion of the cisterna chyli is located in the retroperitoneal space along the right dorsal aspect of the aorta ventral to L1 vertebra, and consists of two small converging and diverging lymphatic vessels. The caudal extent of the cisterna chyli is not included in the study. The thoracic duct lies retroperitoneally along the cranial abdominal aorta between the crura of the diaphragm and forms the cranial continuation of the cisterna chyli. There are two thoracic duct branches. Along T11, T12 and T13, the thoracic duct branches are respectively located along the right and left dorsal aspect of the aorta. Both branches converge, diverge, and are joined by short intermediate branches dorsal to the aorta. Cranial to T10 vertebra, both branches course separately along the descending aorta. Along the dorsal aspect of the left subclavian artery, both thoracic duct branches unite into a single thoracic duct that subsequently joins the cranial mediastinal lymph nodes and insert at the junction of the left external jugular veins with the left subclavian veins. Following the lymphangiogram study, there is a mild increase in the attenuation of the pleural effusion (pre IV contrast mean: 16 HU; post IV contrast mean: 17 HU, post lymphangiogram mean: 37 HU).

CONCLUSION:

NECK:

1. Ventral laryngeal mass with central focus of mineral attenuation, causing partial attenuation of the laryngeal lumen. Differential diagnoses include granulation tissue secondary to prior surgical devocalization or foreign body migration (given the central focus of mineral attenuation) vs. neoplasia (e.g., squamous cell carcinoma, mast cell tumor, lymphoma, rhabdomyosarcoma, chondrosarcoma, oncocytoma, etc.).
2. Mild bilateral medial retropharyngeal lymphadenopathy. Reactive lymphoid hyperplasia is suspected. Metastatic neoplasia is a secondary consideration.
3. Bilateral hypoplasia of the epiphyoid bone.

THORAX:

1. Moderate bilateral pleural effusion and mild to moderate pleuritis/pleural fibrosis, consistent with the recently diagnosed chylothorax. The minimal right sided pneumothorax is consistent with an iatrogenic change secondary to prior thoracentesis.
2. Secondary mild to moderate multilobar pulmonary atelectasis.

3. Mild cranial mediastinal lymphadenopathy with perinodal effusion and/or steatitis, most consistent with underlying reactive lymphoid hyperplasia.
4. Mild esophageal fluid distension, likely secondary to gastroesophageal reflux under general anesthesia.
5. Moderate to severe right-sided chronic renal degeneration including nephrolithiasis.
6. Moderate gallbladder distention (likely secondary to fast), cholecystolithiasis and multifocal hepatic cysts vs. biliary cystadenomas.
7. Moderate gastric distention with fluid and gas, suggestive of gastric stasis (possibly induced by general anesthesia).
8. Mild subcutaneous fluid along the left ventral aspect of the thorax, most consistent with leakage of pleural fluid into the subcutaneous tissues at the time of prior thoracocentesis.

LYMPHANGIOGRAM:

1. Pleural lymphatic leakage as supported by the increased attenuation of the pleural effusion following the lymphangiogram study.
2. Two thoracic duct branches (right dorsal and left dorsal branches - see description above).
3. No evidence of lymphangiectasia.

This report was finalized by Alexandra Beaulieu, DVM, DVSc, DACVR at 2:10 PM Eastern Time on 05/22/2026
Support Email: interviewsupport@vetrad.com
Support Phone: 888.483.8723

 BO NGUYEN

PET OWNER:	NGUYEN	MedVet Silicon Valley	LAB ID:	7201141323
SPECIES:	Canine	7090 Santa Teresa Blvd,	ORDER ID:	323466999
BREED:	Shiba Inu	San Jose, CA 95139	COLLECTION DATE:	5/20/26
GENDER:	Male Neutered	408-649-7070	DATE OF RECEIPT:	5/21/26
AGE:	8 Years	ACCOUNT #: 7918	DATE OF RESULT:	5/22/26
PATIENT ID:	548087	ATTENDING VET: Robin Kelly		

IDEXX Services: **ADD-ON TEST REQUEST 2, Triglycerides, Serum**

Chemistry

5/21/26 (Order Received)
5/22/26 12:36 PM (Last Updated)

TEST	RESULT	REFERENCE VALUE	
Triglyceride	71	20 - 150 mg/dL	

Other

5/21/26 (Order Received)
5/22/26 12:36 PM (Last Updated)

TEST	RESULT
Add-on Test Request 2	Your Add On request has been processed.

 BO NGUYEN

PET OWNER: **THAO NGUYEN**
SPECIES: Canine
BREED: Shiba Inu
GENDER: Male Neutered
AGE: 8 Years
PATIENT ID: 548087

MedVet Silicon Valley
7090 Santa Teresa Blvd,
San Jose, CA 95139
408-649-7070
ACCOUNT #:
ATTENDING VET: Camille Brassard

LAB ID:
ORDER ID: 323831034
DATE OF RECEIPT: **5/22/26**
DATE OF RESULT: **5/22/26**

IDEXX Services: **Coag Dx Analyzer**

Hematology



5/22/26	3:52 PM	3:50 PM	
TEST	RESULT	REFERENCE VALUE	
Citrated Prothrombin Time (PT)	14.0	11.0 - 17.0 seconds	
Citrated Partial Thromboplastin Time (PTT)	121.0	72.0 - 102.0 seconds	H

Patient Name: Nguyen^Bo	Patient ID: 645583	Institution: MedVet Silicon Valley
Patient Birth Date: 05/20/2018	Age: 8	Requester: Camille Brassard
Species: Canine	Sex: MN	Modality: CT
Breed: Shiba Inu	Accession: US798-DR436611	Study Date: 05/22/2026

Final Report

CASE DETAILS:

- History - 1 month history of exercise intolerance, decreased appetite and weight loss. Acute onset of dyspnea: presented to ER yesterday. Pleural effusion consistent with chylothorax. Echo showed hypocontractility, but no structural changes that could explain the chylothorax. The dog is severely lethargic/obtunded and ataxic, which is not typical of an idiopathic chylothorax.
- Lab Findings -
- Clinical Questions to be Addressed in Study -

FINDINGS:

A pre and post IV contrast CT of the abdomen dated 5/22/2026 is available for review. There is a moderate volume of contrast within the urinary bladder at the start of the study.

In the visible portion of the thorax, there is a moderate volume of right sided pleural effusion and mild volume of left-sided pleural effusion. There is mild contrast-enhancement of the parietal pleura bilaterally. There is a 6 mm x 7.8 mm ill-defined nodular region within the right middle lung lobe (series 14 image 13). There is mild wispy soft tissue attenuating regions within the subcutaneous tissues of the left lateral thorax. There is focal thickening of the mid body of the left sixth rib. There is no evidence of peritoneal effusion or peritoneal gas. The gallbladder is moderately distended with ventrally dependent hyperattenuating material. The common bile duct is normal in size and can be traced to its termination at the duodenal papilla. There are multiple small hypoattenuating noncontrast enhancing hepatic nodules (all less than 5 mm in diameter). The spleen is normal with normal contrast enhancement. One of the splenic lymph nodes is mildly enlarged measuring 12 mm x 5.8 mm (series 12 image 226). The adrenal glands are normal. The kidneys are small with irregular margination bilaterally (left 4.3cm, right 4.2cm). There is renal diverticular mineralization bilaterally. There is nephrolithiasis bilaterally (Left: 6 mm x 6 mm x 17 mm, Right: 5.5 mm x 6.2 mm x 6.8 mm and 4.2 mm x 3.1 mm x 7 mm). Renal pelvis dilation is not identified. The ureters can be traced to their termination at the trigone. The urinary bladder contains a moderate amount of hyperattenuating contrast at the start of the study limiting evaluation of luminal contents. The visible urinary bladder wall is unremarkable. There are multiple regions of mineralization within the prostate. The prostate is normal in size measuring 1.7 cm x 1.4 cm. These regions of mineralization connect with the contrast column from within the urethra on the postcontrast series. The urethra is otherwise normal. The pancreas is normal. The gastric lumen contains a moderate amount of gas and mild amount of fluid. The small intestines contain a mild amount of gas and fluid. The colon contains gas. One of the jejunal lymph nodes is mildly to moderately enlarged (11 mm x 11 mm, series 15 image 314). The lumbar spinal column is well aligned. There is a mild to moderate amount of hyperattenuating material in the ventral to mildly left-sided spinal canal at the T13-L1 causing mild to moderate lifting and compression of the spinal cord at this site. There is mild hyperattenuating thickening of the dorsal annulus at L2-L3 compression of the spinal cord. There is moderate thickening of the femoral necks with osteophyte formation bilaterally. There is flattening and sclerosis of the acetabulum with osteophyte formation bilaterally.

CONCLUSION:

1. Mineralization within the prostate. Possible differentials include dystrophic mineralization, retained contrast within prostatic invaginations, or emerging neoplasia. BRAF testing could be considered.
2. Mild to moderate enlargement of one of the jejunal lymph nodes and mild enlargement of one of the splenic lymph nodes. Possible differentials include reactive or less likely metastatic lymphadenopathy.
3. Moderate bilateral renal degenerative changes with nephrolithiasis indicating chronic renal disease. Clinical significance be correlated with chemistry and urinalysis findings.
4. Multiple hypoattenuating hepatic nodules. Benign etiologies (e.g. nodular regeneration, hepatic cyst) are prioritized.
5. Moderate gallbladder distention with gravity-dependent hyperattenuating material. This likely represents cholecystolithiasis or mucinous hyperplasia.
6. Chronic intervertebral disc disease at T13-L1 moderate spinal cord compression. Chronic intervertebral disc disease at L2-L3 with minimal spinal cord compression. Clinical significance to be correlated to neurologic exam findings.
7. Bilateral pleural effusion with evidence of pleuritis/pleural fibrosis. This is consistent with the previously diagnosed chylothorax.
8. Ill-defined nodular region within the right middle lung lobe likely representing atelectasis as it is not identified on the previous days CT. A true pulmonary nodule is considered less likely.
9. Edema within the left lateral thoracic tissues. Possible differentials include hemorrhage/hematoma formation or leakage of pleural fluid into the subcutaneous tissues following thoracocentesis.
10. Hip dysplasia with moderate osteoarthritis of the coxofemoral joints bilaterally.
11. Healed fracture of the left sixth rib.

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This report was finalized by Dalal Hares, DVM, DACVR at 1:10 PM Mountain Time on 05/24/2026
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MedVet Silicon Valley**7090 Santa Teresa Blvd.****San Jose, CA 95139****Report No.****26052697****Phone: (408) 649-7070**Name **Bo US798-DR436381**Sp, Breed **K9, Shiba Inu**Collected **5/21/26**Owner **Nguyen, Thao**Sex **MN**Received **5/22/26**Doctor **Kelly**DOB/Age **05/20/2018**Reported **5/24/26**

HISTORY

Acute onset dyspnea/lethargy with inspiratory effort and stridor on inspiration. Soft tissue mass on larynx, irregular and friable.

SPECIMEN

Multiple sections of tissue are placed in cassette A.

MICROSCOPIC DESCRIPTION

Larynx: Three, irregularly sized tissue specimens are examined in duplicate. The mucosal epithelium has papillary hyperplasia, erosion and ulceration. The lamina propria is focally replaced by inflamed granulation tissue and necrosis. Inflammation consists primarily of neutrophils with paving in small vessels. An exudate of mucus and minimal numbers of neutrophils is present.

DIAGNOSES

Larynx: laryngitis, necrosuppurative, ulcerative, subacute and severe with granulation tissue and papillary epithelial hyperplasia

COMMENT

The cause of these inflammatory and degenerative changes in the laryngeal mucosa is not determined. Although no evidence of neoplasia is identified here, this finding does not exclude the possibility of neoplasia on a deeper (unsampled) tissue plane.

PATHOLOGIST

Brian Murphy, DVM, PhD

Diplomate, American College of

Veterinary Pathologists

Veterinarians: For consultation on this report, questions or to request additional testing, please e-mail

CustomerService@vdxpathology.com or call (530) 753-4285

Full Biopsy - 1 Tissue